Digital-camera exposure for film scanning

Why use Camera-41_exp?

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Camera-41_exp is a standalone tool for choosing the digital-camera shutter time when photographing negatives, at a fixed selected aperture.

Avoid clipping in the RAW capture

An empty frame containing film base plus fog (FB+F) normally transmits more light than denser image areas of a negative. It is therefore an important exposure reference. If the shutter time is too long, one colour channel may reach its limit. Measurable information is then lost; making the preview darker cannot restore it.
Exposure reads the original Bayer values and compares signal and headroom relative to LibRaw's linearity limit in R, G1, G2 and B. The aim is a strong usable signal with margin, not the brightest image on screen.

What the app does

It compares RAW files at different shutter times, normally chooses an area away from the edges and plots signal and clipping margin. It proposes the longest actually measured time that meets its criteria. A proposal may need further test captures.

What it does not do

Camera-41_exp does not measure dark noise or full S/N and does not create a production qualification. It does not control the camera. The exposure choice concerns the digital capture, not the original analog exposure when the film was photographed.

Quick start

1. Open start.m in MATLAB and press Run, or run startCamera41_exp; from the app folder.
2. Add RAW files of the same empty FB+F sample at different shutter times.
3. Describe the light and sample; confirm that all files show the same sample.
4. Press Analyse RAW images. Review the report and Exposure comparison.
5. Select a measured passing time, set the camera yourself and confirm with a fresh capture.

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Installation and startup

Camera-41_exp v0.1.4 runs in MATLAB. The app checks its own isolated Python 3.13.6 environment at startup. See PYTHON_ENVIRONMENT.md for setup and repair. PRE and NEG do not need to be installed.

1. Download and extract

Follow the GitHub link in the footer and select Releases > v0.1.4. Download Camera-41_exp-v0.1.4-macos.zip and extract the entire archive into a local writable folder. Keep src, tools, VERSION and the other supplied files together; do not move only start.m.

2. Start in MATLAB

Open MATLAB, open start.m from the extracted app folder and press Run. Alternatively, set that folder as MATLAB's Current Folder and enter startCamera41_exp; in the Command Window. The main application window opens.

Requirements and supplied RAW reader

Verified environment: macOS on Apple Silicon with MATLAB R2025b. Image Processing Toolbox is needed for manual region editing with drawrectangle. MATLAB requires its own licence. The supplied LibRaw reader is selected for the Mac architecture. No Homebrew installation is needed in the verified environment. An Intel reader and libraries are included, but MATLAB operation on a physical Intel Mac has not been verified. Windows, Linux and older MATLAB releases are not qualified for this release.

After installation or an update

No sample photographs are included. Add your own files with Add FB+F RAW files. The log is created at logs/camera41_exp.jsonl inside the app folder; use Log > Open complete log. Close the Camera-41_exp window before replacing program files, then restart the app. If macOS blocks the downloaded reader, inspect the security message and local log.

Check the download

The release page includes SHA256SUMS.txt. On macOS, compare a file using shasum -a 256 followed by its filename. RELEASE_MANIFEST.json inside the ZIP records the source commit and per-file checksums. INSTALL.md also contains the full installation instructions.

Native RAW reader

The LibRaw reader is validated against the physical ARM/Intel CPU, including its unsuffixed fallback. A wrong-architecture binary is rejected. Local rebuilding requires matching native LibRaw and libomp libraries. Physical Intel operation remains unverified.

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Capture files and select an area

Use an empty film frame containing base plus fog from the film being scanned. A negative with a scene, text, perforations or a holder edge is not a suitable empty FB+F area.

Keep the setup fixed

Use the same camera, ISO, aperture, light mode and geometry throughout the bracket. Choose the aperture first and change only shutter time. Start a separate comparison if the light, film sample or setup changes. Light source / mode / setup and Physical film base + fog identification document what was measured.

Add and preview RAW files

Add FB+F RAW files accepts one or more files. Later selections are added to the list; the same path is not duplicated. The file chooser remembers the latest RAW folder across app restarts. Remove selected removes a row from the bracket, not the original file.
Select a row and press Preview selected RAW to inspect it. RAW preview is a display image with brightness adjustment; screen brightness does not determine whether the measurement passes.

Automatic area

Automatic area is enabled by default. The central 70 percent of visible width and height is used, leaving a 15 percent border on each side. The same area is measured throughout the bracket. The app does not hunt for a small favourable patch to make an excessive exposure pass.
A per-Bayer-channel screen checks for substantial structure, large illumination differences or insufficient signal. This assists inspection but cannot reliably identify the physical contents of a film frame.

When a warning requires adjustment

If a substantial deviation is detected, the affected image is shown with the proposed area marked in yellow. Read the warning, then move or resize the rectangle onto clean empty film. Adjust marked area enables manual adjustment. X, Y, Width and Height refer to native visible sensor pixels. The rectangle is constrained to the image bounds.
Do not exclude a genuinely clipped bright part of the FB+F sample merely to obtain PASS. If the correct empty sample cannot be identified, capture a new test frame.

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Interpret the results

Successful RAW reading and a passing exposure are different outcomes. FAIL can be a valid measurement from a file that was read successfully.

PASS, REVIEW and FAIL

PASS: every channel retains at least 1/3 EV headroom at the 99.9th percentile, with too few values at/above the linearity limit to trigger FAIL.
REVIEW: no channel triggers FAIL, but at least one retains less than 1/3 EV margin. Do not increase exposure; try a slightly shorter time.
FAIL: at least one channel has 0.01 percent or more of its measured values at/above the limit. First check the sample and measurement area. If they are correct, try a shorter shutter time.
The reference is the RAW linearity limit after black subtraction, not a JPEG histogram. Loss of linearity can matter before hard sensor saturation.

Plots and table

Exposure comparison plots RAW signal codes and headroom in EV against shutter time, separately for R, G1, G2 and B. A single image gives only one point per channel; add other times for a comparison.
The table shows file, time, status, minimum headroom and the largest channel fraction at/above the limit.
The text report provides per-channel measurements and explanations. Positive margin means headroom below the limit; negative margin means the percentile value exceeds it.

A candidate may need more evidence

No passing exposure: try shorter times if the correct FB+F area clips.
Longest passing time without a higher non-passing capture: extend the bracket upward to bound the choice.
Too wide a gap to the next higher non-passing time: test an intermediate time.
The internal proportionality check needs at least three distinct passing times spanning at least one stop (2× time). Less evidence means an insufficient check, not automatic proof of a faulty camera.

Choose a time with margin

The list at the bottom contains measured passing times. The app's candidate is preselected. Confirm the chosen time with a fresh FB+F capture before a longer scanning session. Do not lengthen exposure simply because the preview looks better. Revisit the test if camera settings, lighting or film base change.

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After the test, logging and help

Camera-41_exp works independently. No JSON file is transferred from Camera-41_exp to PRE or NEG, and neither program needs to be open to use it.

Continue in PRE with current captures

Set the digital camera to the chosen time yourself. Capture the current film's empty FB+F frame and two separate dark RAWs with the same camera, ISO, shutter time and aperture. For the dark frames, block light from reaching the sensor and use relevant matching capture conditions.
Load those files directly in PRE tab 7. FB+F and the dark pair can be inspected with a common sensor area and used for noise and S/N measurement. PRE includes its measured result in the film characterisation delivered to NEG. Camera-41_exp does not export that measurement and need not be installed for PRE to work.

If something looks wrong

Nothing in RAW preview: select a file and press Preview selected RAW. Plots are in the other tab, Exposure comparison.
No selectable time: read the channel report. Every file may be REVIEW or FAIL. One failed capture cannot establish a useful time.
RAW file missing: check that the file or external drive is still available.
Slow chooser or an error: inspect the log. Preparation, dialog, file registration and analysis are recorded separately.
After a code update: close the Camera-41_exp window and start the app again.

Log and retained information

Log > Open complete log opens logs/camera41_exp.jsonl in the app folder. Each record contains UTC time, process ID, event and details. File selections, settings, measurements, durations and errors are logged. Native chooser duration includes the time you spend selecting files; it does not by itself identify when the dialog first became visible.
The latest RAW folder is stored in MATLAB preferences. The file list and other window settings are not restored after closing. Original RAWs remain unchanged. The log is local and excluded from Git.

Requirements, edition and licences

This handbook covers Camera-41_exp v0.1.4, 27 October 2026. MATLAB and the supplied LibRaw reader are required; the Apple Silicon reader has been tested on the development computer. On another computer the reader may need rebuilding with LibRaw and pkg-config.
The project's own code and documentation use GPL-3.0-or-later. See LICENSE and LICENSE_NOTICE.md. Third-party notices for LibRaw, JPEG-turbo, libomp and Little CMS are in third_party/licenses. MATLAB requires its own licence.

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